MILITARY SPECIFICATION

HOSE ASSEMBLY, RUBBER, LIGHTWEIGHT, MEDIUM PRESSURE, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for hose assemblies suitable for aircraft fuel and oil systems within the limits specified herein.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-H-5606	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, And Ordnance
MIL-L-7808	Lubrication Oil, Aircraft Turbine Engine, Synthetic Base
MIL-H-83796/1	Hose Assembly, Rubber, Lightweight, Medium Pressure, Field Attachable, Flare To Flare
MIL-H-83796/2	Hose Assembly, Rubber, Lightweight, Medium Pressure, Field Attachable, Flare To Flare, With Lockwire Hole
MIL-H-83796/3	Hose Assembly, Rubber, Lightweight, Medium Pressure, Field Attachable, Flareless To Flareless
MIL-H-83796/4	Hose Assembly, Rubber, Lightweight, Medium Pressure, Field Attachable, Flareless To Flareless, With Lockwire Hole
MIL-H-83796/5	Hose Assembly, Rubber, Lightweight, Medium Pressure, Field Attachable, Flare To Flange
MIL-H-83796/6	Hose Assembly, Rubber, Lightweight, Medium Pressure, Field Attachable, Flare To Flange, With Lockwire Hole
MIL-H-83796/7	Hose Assembly, Rubber, Lightweight, Medium Pressure, Field Attachable, Flareless To Flange

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MIL-H-83796/8	Hose Assembly, Rubber, Lightweight, Medium Pressure, Field Attachable, Flareless to Flange, With Lockwire Hole
MIL-H-83796/9	Hose Assembly, Rubber, Lightweight, Medium Pressure, Field Attachable, Flange To Flange.
MIL-H-83797	Hose, Rubber, Lightweight, Medium Pressure, General Specification For
MIL-F-83798	Fittings, Rubber Hose, Lightweight, Medium Pressure, General Specification For
MIL-F-83798/1	Fittings, Rubber Hose, Lightweight, Medium Pressure, Flare, Swivel Nut, Straight
MIL-F-83798/2	Fittings, Rubber Hose, Lightweight, Medium Pressure, Flared, Swivel Nut, 45 Degrees
MIL-F-83798/3	Fittings, Rubber Hose, Lightweight, Medium Pressure, Flared, Swivel Nut, 90 Degrees
MIL-F-83798/4	Fittings, Rubber Hose, Lightweight, Medium Pressure, Flareless, Swivel Nut, Straight
MIL-F-83798/5	Fittings, Rubber Hose, Lightweight, Medium Pressure, Flareless, Swivel Nut, 45 Degrees
MIL-F-83798/6	Fittings, Rubber Hose, Lightweight, Medium Pressure, Flareless, Swivel Nut, 90 Degrees
MIL-F-83798/7	Fittings, Rubber Hose, Lightweight, Medium Pressure, Swivel Flange, Straight
MIL-F-83798/8	Fittings, Rubber Hose, Lightweight, Medium Pressure, Swivel Flange, 45 Degrees
MIL-F-83798/9	Fittings, Rubber Hose, Lightweight, Medium Pressure, Swivel Flange, 90 Degrees

STANDARDS

Federal

FED-STD-601	Rubber: Sampling And Testing

Military

MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
MIL-STD-1523	Age Controls Of Age-Sensitive Elastomeric Materials
MS20756	Flange, Swivel, Retaining

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PUBLICATIONS

Defense Supply Agency Cataloging Handbook

H4-1 (SB 708-41)	Federal Supply Code For Manufacturers - Name To Code

Federal Aviation Agency Technical Standard Order

TSO-C53a	Fuel And Oil System Hose Assemblies

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Society of Automotive Engineers Specifications

ARP 611	Tetrafluoroethylene Hose Assembly Cleaning Methods
ARP 1055	Fire Resistance and Fire Test Requirements for Fluid System Components

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 2 Pennsylvania Plaza, New York NY 10001.)

3. REQUIREMENTS

3.1 General. The hose assemblies furnished under this specification shall consist of hose conforming to MIL-H-83797 and end fittings conforming to MIL-F-83798 which, when assembled, meet the quality assurance provisions of Section 4 herein.

3.2 Design and construction. The design and construction of the hose assembly shall conform to MIL-H-83796/1 thru MIL-H-83796/9 as applicable (see 6.2a).

3.3 Selection of specifications and standards. Specifications and standards for parts, materials, procedures, and processes not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Performance. The hose assemblies shall meet the requirements listed below when subjected to the applicable tests in Section 4:

		Paragraph
a.	Coupling	4.4.2
b.	Proof pressure	4.4.3
c.	Leakage	4.4.4
d.	Burst pressure	4.4.5
e.	Fire resistance (when required see 6.2b)	4.4.6

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3.5 Identification of products. Products shall be identified in accordance with MIL-STD-130. In addition, the following special marking shall be required.

3.5.1 Fittings. Fittings shall be permanently marked with:

a. Manufacturer's name or trademark.

b. Part number of the fitting, including dash size, (see MIL-F-83798/1 thru MIL-F-83798/9).

3.5.2 Hose assembly. A permanent band located at one end of assembly, shall provide:

a. Assembler's name or trademark.

b. Part number, including dash size, formulated as shown on MIL-H-83796/1 thru MIL-H-83796/9.

c. Operating pressure in pounds per square inch (psi).

d. Date of assembly in quarter of year and year:

Example: A3Q74
Quarter — Year

e. Hose manufacturer's Federal Supply Code in accordance with Handbook H4-1.

f. Date of hose manufacture in quarter of year and year.

g. ARP 1055 (when applicable).

3.6 Age control. Hose assemblies shall not exceed the age limits established in MIL-STD-1523.

3.7 Workmanship. Workmanship shall be of the quality necessary to produce hose assemblies free from all defects which affect proper functioning in service. All products shall be free from oil, grease, dirt, or other foreign material both internally and externally consistent with good manufacturing practices. Cleaning shall be in accordance with class 0 of ARP 611.

4. QUALITY ASSURNACE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of tests. The inspection and testing specified herein are classified as follows:

a. Individual (see 4.2.1).

b. Sampling and periodic control (see 4.2.2).

4.2.1 Individual. Each hose assembly submitted for acceptance shall be subjected to the following tests:

a. Examination of product (4.4.1).

b. Proof pressure (4.4.3).

4.2.2 Sampling and periodic control. The following tests shall be performed on at least one hose assembly taken at random from each lot. Failure of any sample to withstand the specified tests shall be cause for rejecting the lot. A lot is defined as 500 assemblies, or fraction thereof, of a given dash size.

a. Coupling (4.4.2).

b. Proof pressure (4.4.3).

c. Leakage (4.4.4).

d. Burst pressure (4.4.5).

4.2.3 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and corrected. The contractor shall explain to the Government representative the cause of the failure and the action taken to preclude recurrence. After correction, all of the quality conformance tests shall be repeated.

4.2.3.1 Individual tests may continue. For production reasons, individual tests or other quality conformance tests may be continued pending the investigation of the test which failed. Final acceptance of items on hand or produced later, shall not be made until it is determined that all items meet all the requirements of this specification.

4.2.3.2 Defects in items already accepted. If the investigation of quality conformance test failure indicates that defects may exist in items already accepted, the contractor shall fully advise the procuring activity of all defects liable to be found and the method of correcting them.

4.3 Test conditions. Unless otherwise specified herein the test procedures shall be in accordance with FED-STD-601 except that the test fluid shall conform to MIL-H-5606 or water when specified. Hoses must be cleaned and dried after test.

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4.4 Test methods.

4.4.1 Examination of product. Hose assemblies shall be examined to determine conformance with this specification in regard to the following, as applicable:

a. Size

b. Finish

c. Dimensions

d. Identification

e. Workmanship

f. Cleaning

4.4.2 Coupling. The measurement of the bulging of hose inner tubes caused by attachment of the fitting shall be made with a ball-end type gage. The diameter of the ball shall be within 0.001-inch of the minimum bulge diameter specified on Figure 1. The weight of each gage in ounces shall be equal to the dash number of the fitting being tested. In taking the measurement, the gage shall be placed inside the end of the assembly without lubrication, and without pushing through. The gage shall fall through the bulge under its own weight. This test shall be made only on straight fittings.

4.4.3 Proof pressure. The hose assemblies shall be subjected to the proof pressure of Table I for not less than 30 seconds and not more than 5 minutes, using oil conforming to MIL-L-7808 or hydraulic fluid conforming to MIL-H-5606, or water. There shall be no evidence of leakage.

4.4.4 Leakage. The hose assemblies shall be subjected to 70 percent of the minimum burst pressure specified in Table I for 5 minutes, using oil conforming to MIL-L-7808. Th pressure shall then be released to 0 psi and then reapplied to 70 percent of minimum burst pressure and held for an additional 5 minutes. There shall be no leakage from the fitting, no seepage through the hose, or other malfunction.

4.4.5 Burst pressure. The hose assemblies shall be pressurized until failure, using MIL-L-7808 oil, MIL-H-5606 hydraulic fluid, or water. The rate of pressure rise shall be 25,000 psi, +0 psi, -10,000 psi per minute. There shall be no leakage, burst, or fitting blow-off below minimum burst pressure specified in Table I. The type failure and pressure at which failure occurred shall be recorded.

4.4.6 Fire resistance. When fire-resistance supplemental tests are required (see 6.2b), the hose assemblies shall be tested and shall meet the requirements of ARP 1055, type 1, class A. Hose assemblies approved to the fire test requirements of TSO-C53a prior to the effective date of this specification will be considered in compliance with this requirement. A protective sleeve over the hose or fitting may be required to facilitate compliance with this requirement.

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TABLE I

PHYSICAL REQUIREMENTS OF HOSE, FITTINGS, AND HOSE ASSEMBLIES

Hose Size	Operating Pressure, psi (max)	Proof 1/ Pressure, psi	Burst Pressure, psi (min)	Operating Temperature (fluid or ambient)
-3	1,000	3,000	6,000	-65°F to +250°F
-4	1,000	3,000	6,000	-65°F to +250°F
-5	1,000	3,000	6,000	-65°F to +250°F
-6	1,000	3,000	6,000	-65°F to +250°F
-8	1,000	2,500	5,000	-65°F to +250°F
-10	1,000	2,500	5,000	-65°F to +250°F
-12	1,000	2,000	4,000	-65°F to+250°F
-16	750	1,500	3,000	-65°F to +250°F
-20	500	1,300	2,000	-65°F to +250°F
-24	250	800	1,750	-65°F to +250°F
-32	200	600	1,200	-65°F to +250°F

1/ Assemblies having aluminum flange fittings shall be pressure tested at the rated proof pressure or 1,500 psi, whichever is less (see MS 20756).

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4.5 Inspection of preparation for delivery. The preservation, packaging, packing and marking of hose, fittings and hose assemblies shall be examined to determine conformance with Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with the applicable packaging standard or packaging data sheet for the desired level of protection specified by the procuring activity (see 6.2c).

6. NOTES

6.1 Intended use. The hose assemblies are intended for use in aircraft fuel and oil systems at the pressures and temperatures specified in Table I herein. The products are not recommended for vacuum or hydraulic impulse application.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and applicable specification sheet.

b. Whether supplemental tests are required (see 4.4.6).

c. Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 Supersession data. This specification supersedes MIL-H-58089(AV) dated 10 March 1969. In addition, MIL-H-83796/1 and MIL-H-83796/2 supersede MS87028(AV) dated 07 June 1971; MIL-H-83796/3 and MIL-H-83796/4 supersede MS87029(AV) dated 07 June 1971; MIL-H-83796/5 and MIL-H-83796/6 supersede MS87030(AV) dated 07 June 1971; MIL-H-83796/7 and MIL-H-83796/8 supersede MS87031(AV) dated 07 June 1971; MIL-H-83796/9 supersedes MS87032(AV) dated 07 June 1971.

Custodians:
Army - AV
Air Force - 82

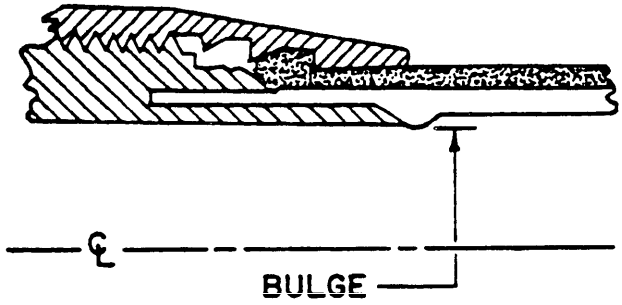
Review Activity:
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Hose Size	Minimum Bulge Diameter	Hose Size	Minimum Bulge Diameter
-3	.094	-10	.469
-4	.141	-12	.563
-5	.203	-16	.750
-6	.266	-20	1.000
-8	.344	-24	1.250
		-32	1.625

FIGURE 1. Coupling Bulge

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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